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Roll No

EC-3005-CBGS

B.E. III Semester

Examination, December 2020

Choice Based Grading System (CBGS)

Measurements and Instrumentation

Time : Three Hours

Maximum Marks : 70

Note: i) Attempt any five questions.

ii) All questions carry equal marks.

iii) Sketch neat diagram, if any.

1. a) Explain the following:
 - i) Accuracy and Precision
 - ii) Sensitivity
 - iii) Resolution
- b) Explain the construction and working of chopper type DC voltmeter.
2. a) A voltmeter having a sensitivity of $10\text{k}\Omega/\text{V}$ reads 180V on a 200V scale when connected across an unknown resistor. The current through the resistor is 2mA. Calculate the percentage of error due to loading effect.
- b) Give the comparison between Dual trace and Dual beam oscilloscopes.
3. a) With the help of a suitable diagram explain the function of time base generator in a CRO.

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- b) Explain the following controls of a CRO :
 - i) Intensity
 - ii) Focus
- 4. a) Explain the principle working of dual trace and dual beam CRO.
b) Discuss the working of storage CRO.
- 5. a) Compare analog and digital instruments.
b) Describe the working Digital voltmeters and multimeter.
- 6. a) Define rectifiers in terms of AC voltmeter and chopper type DC voltmeter.
b) Describe different types of sweeps used in CRO.
- 7. a) Write short notes on impedance Measurement of Q meter.
b) Briefly describe measurements of any two non-electrical quantities.
- 8. Write short notes on:
 - a) Photo-transistor
 - b) Sensitivity of digital multimeter
 - c) Time base circuit
